MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION OCTOBER 2025

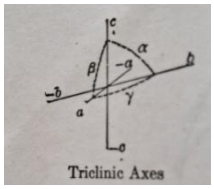
THIRD SEMESTER

DISCIPLINE SPECIFIC COURSE (DSC) - MG3DSCGEO200- MINERALOGY

ANSWER SCHEME

PART A

SHORT ANSWER IN 60 WORDS
ANSWER ANY **7** QUESTIONS
EACH QUESTION CARRIES **2** MARKS

1. 

Triclinic Axes

2. The slight attraction whereby mineral moves, or tends to move, towards regions of stronger magnetic field strength. This tendency varies directly with the strength of the applied magnetic field. **Definition -(1.5 marks), Examples-(0.5 mark)**
3. Eye piece, Objective lens, Stage, Polarizer, Analyzer, Condenser, Mirror, Bertrand lens etc.- **(Any four- 2 marks)**
4. Anisotropic minerals show double refraction. -(**1 mark)** Examples- Calcite, Quartz, Mica. - **(1 mark)**
5. Diamond and Graphite -**(2 marks)**
6. The sulphide group of minerals is predominantly metallic in character, and many of these minerals are opaque. Their hardness ranges from about 1 to 6 or more, as in the case of pyrite. Definition-**(1 mark)**. A good example of a sulphide mineral is galena (PbS), the most important lead-bearing mineral in the Earth's crust. Galena is also one of the most common sulphide minerals and serves as the primary source of lead. -**Example (1 mark)**
7. Amphiboles contain essential hydroxyl (OH) groups in their crystal structure, while pyroxene does not. -**(2 marks)**
8. Hematite (Fe_2O_3) and Magnetite (Fe_3O_4). -**(2 marks)**

PART B

SHORT NOTES IN 250 WORDS

ANSWER ANY **3** QUESTIONS

EACH QUESTION CARRIES **8** MARKS

9. The color of powder of a mineral as obtained by scratching the surface of the mineral with a knife or file, if the mineral is not too hard, by rubbing it on an unglazed porcelain surface, is called the streak. **Definition- (2 marks)**. The streak is often very important quality in distinguishing minerals. Streak is an independent property while compared to color. For example- hematite and magnetite are black in color, while their streak cherry red and black in color**. - Explanation (4 marks). Example (2 marks)**
10. Rotation symmetry: a symmetry operation where an object is rotated around an axis by specific angle and returns to its original position. **(4 marks)**

    Rotoinversion: A combined symmetry operations involving a rotation followed by an inversion through a center point. **-(4 marks)**
11. When the ray of light coincides with the perpendicular no change of direction takes place, the ray proceeds onwards into the second medium without deviation but with a change of velocity. **Definition- (3 marks)**. The phenomenon of total internal reflection take advantage of in cutting the gemstone (Diamond). **Significance- (2 marks). Example-(1 mark). Figure. -(2 marks)**
12. Halides find their greatest use in the production of sodium compounds and chlorine. Sylvite and other potassium salts are used in fertilizers, as well as in the manufacture of dyes and soaps. **Definition with examples- (3.5 marks).** Large quantities of fluorite are used in metallurgical operations as a flux. Halides are also essential for making plastics such as polyvinyl chloride (PVC) and synthetic rubber. In electronics, halides play an important role in the production of semiconductors and are used in processes like etching and metal coating. They also help control electrochemical reactions and prevent corrosion in various industries. **– Uses-(4.5 marks).**
13. Trigonal planar coordination is simple, flat, and involves only three ligands arranged around a central atom in a single plane. Cuboctahedral coordination is complex, three-dimensional, and highly symmetrical, accommodating twelve ligands surrounding the

central atom in a tightly packed polyhedral arrangement. **Explanation (5 marks). Figure -(3 marks)**

PART C

ESSAYS IN 400 WORDS
ANSWER ANY **1** QUESTION
EACH QUESTION CARRIES **12** MARKS

14. Explanation about quartz wedge, gypsum plate and mica plate. **(6 marks)**. Quartz wedge- offers variable retardation, helps in determination of birefringence. Gypsum plate- helps in determination of optical sign. Mica plate- Gives low retardation and it is ideal for weakly birefringence mineral- **(3 marks). Applications (3 marks).**
15. Definition of polymorphism- **(2 marks).** Detailed explanation of Aluminium silicate group- Andalusite, Sillimanite, Kyanite-**(6 marks). Significance-(2 marks) Figure-(2 marks)**